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LinguaAI: Speech-to-Text & AI-Powered Endangered Language Preservation Platform

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Introduction

- Endangered languages face imminent extinction due to declining native speakers, globalization, and a severe lack of digital preservation tools.
- LinguaAI is an AI-assisted web platform designed for automated speech-to-text transcription, translation, and digital archiving of endangered and regional languages.
- The platform integrates real-time Web Speech API recognition, custom language database management, analytics dashboards, and interactive admin control panel.

Background and Motivation

- Over 3,000 endangered languages worldwide lack digital documentation, leading to irreversible loss of cultural heritage and linguistic diversity.
- Traditional documentation relies on manual field notes and audio recordings, which are slow, inaccessible, and difficult to search or analyze.
- Existing speech-to-text tools primarily target dominant global languages (English, Spanish, Mandarin), completely neglecting minority and regional dialects.
- Motivation: Build an accessible, real-time Web-based platform to transcribe, catalog, manage, and preserve endangered languages with administrative oversight and analytics.

Relevance and Importance

- New Insight: A unified web architecture combining browser-level Web Speech recognition with MySQL relational indexing for real-time multilingual transcription archiving.
- Relevant to: Linguists, cultural researchers, educational institutions, language revival groups, and government cultural departments.
- Why it matters: Digital preservation enables real-time acoustic transcription, searchable transcript archives, language status tracking, and secure role-based administrative control.

Literature Review

- Traditional Linguistic Archiving (e.g., ELAR, DOBES): Focuses on static audio/video storage; lacks real-time automated transcription and web accessibility [1].
- Commercial Automatic Speech Recognition (ASR) (Google Cloud, IBM Watson): High accuracy for major languages, but limited API support for low-resource endangered languages [2].
- Web Speech API & Browser-Based ASR: Provides zero-latency, cross-platform voice recognition via BCP-47 language tags directly in browser environments [3][4].
- Database & Analytics Architectures for Linguistics: Relational schemas (MySQL) combined with real-time character/word count indexing optimize query speeds for speech corpora
- Research gap: Existing platforms either lack web-based real-time speech transcription or lack integrated administrative management for endangered language status tracking.

Objectives

- Design and implement a responsive web application for real-time speech-to-text transcription.
- Create a centralized MySQL database schema to store users, endangered language metadata, transcriptions, and messages.
- Integrate Web Speech API supporting dynamic BCP-47 language selection and active status toggling.
- Implement role-based access control with secure authentication and admin dashboard management.
- Provide real-time analytics, character/word metrics, and CSV data export capabilities.
- Deploy and test the system across multiple browsers for latency, accuracy, and accessibility.

Required Tools & Technology

- Programming Language: JavaScript (ES6+), PHP 8.x, HTML5, CSS3
- Database: MySQL / MariaDB (Relational schema with language & transcription indexing)
- Speech & Web APIs: Web Speech API (SpeechRecognition engine, BCP-47 language codes)
- Server & Environment: XAMPP / Apache Web Server, AJAX / Fetch API
- Development & Version Control: VS Code, Git & GitHub, Node.js
- Deployment & Export: Local XAMPP Web Deployment, CSV Data Exporter, System Analytics Engine

Method/ Approach

- **System Architecture:** Client-server architecture with PHP backend processing AJAX requests and interfacing with MySQL database.
- **Speech Processing Pipeline:** Capture microphone input via Web Speech API, perform real-time speech-to-text tokenization, and compute word/character metrics.
- **Database Schema Design:** Define relational tables for users, languages, recognition_languages, transcriptions, and messages with foreign key constraints.
- **Admin Panel & Analytics:** Implement secure admin login, language status management, transcript review, and system usage analytics.
- **Security & Validation:** Enforce input sanitization, password hashing, session isolation, and structured error logging.
- **Testing & Deployment:** Conduct unit testing (TC-01 to TC-06), AJAX latency testing, and local XAMPP deployment.

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